

Krishnaa Vadivel | Electrical Engineer

| Semiconductor Devices | Photonics | FPGA Systems

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Summary

Electrical engineering PhD researcher with Cornell and Yale ECE training across semiconductor physics, embedded systems, photonics, and accelerator-enabled scientific computing. Experience spans FPGA and MCU development, CUDA/OpenGL engineering tools, semiconductor-device modeling, and digital-design projects in Verilog.

Experience

Yale University, Electrical Engineering

PhD Research Assistant / Embedded Systems TA

2021–present

- Conduct research at the intersection of semiconductor physics, condensed-matter modeling, and high-performance scientific computing, with supporting coursework and theory development in semiconductor devices, photonics, and signals/systems.
- Teach and support embedded systems instruction, reinforcing firmware, instrumentation, debugging, and system-integration skills relevant to practical EE development.
- Build numerical and software workflows for device-scale and materials-scale simulation using Python, C++, HPC toolchains, and accelerator-enabled platforms.

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed embedded firmware and FPGA-based sensor-logging functionality for downhole geological tools using Lattice FPGA, PIC32, dsPIC, STM32, and C8051-class platforms.
- Built CUDA/OpenGL software for large acoustic waveform processing, engineering visualization, and diagnostic analysis in field-tool workflows.
- Supported circuit and system bring-up with oscilloscope-based validation, field testing, and targeted PCB updates in Altium.

FindLight

Photonics Technical Writer

2018–2019

- Wrote technical content on lasers, integrated optics, and optoelectronic components for engineering-facing audiences.

Selected Projects

FPGA AI Accelerator: Small Verilog matrix-multiply accelerator with simulation-backed verification for ML-oriented datapath design.

Semiconductor Simulation: Numerical modeling of semiconductor electrostatics, drift-diffusion transport, and pn-junction-style device behavior.

Integrated Photonics Modeling: FDTD simulation of resonator and band-pass style photonic structures using Meep.

Digital Design: Single-cycle MIPS processor in Verilog with self-checking test bench and documentation.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University*MS, Electrical Engineering**2023***Cornell University***MEng, Electrical and Computer Engineering**2017–2018***Cornell University***BS, Electrical and Computer Engineering**2013–2017*

Technical Skills

Hardware: Verilog, FPGA, MCU firmware, PCB bring-up, instrumentation, oscilloscope-based debugging**Devices:** Semiconductor physics, pn junctions, MOSFET concepts, transport modeling, thin films**Photonics:** Integrated photonics, lasers, optics, FDTD simulation**Software:** Python, C, C++, CUDA, OpenGL, Linux**Systems:** Docker, Kubernetes, scientific computing, HPC workflows